



BEFORE

Over millions of years, Earth has experienced a range of temperatures and climatic conditions that have played a part in the extinction or survival of whole groups of species, and changed the face of the planet.

THE ICE AGES

There is geological evidence (seen in rock surfaces and textures) for **four major ices ages** in Earth's history. The earliest of these is believed to have occurred around 2.7 to 2.3 billion years ago during the Proterozoic period.

HOT PLANET

Temperatures in the past were **generally far higher than today**. Following the extinction of dinosaurs about 65 million years ago, perhaps due to climatic change, average temperatures rose to about 82°F (28°C). Tropical rainforests proliferated on Earth.

THE BIG CHILL

The **abrupt cooling about 1.5 million years ago** that led to the last Ice Age, known as the **Pleistocene epoch**, probably resulted from small shifts in Earth's tilt toward the Sun.

The Ice Age

Much of human history unfolded during the dramatic climatic shifts of the most recent Ice Age, which began about 1.5 million years ago. Our ability to adapt to changes in climate has been crucial to the development of civilization but, adversely, may be the cause of future global warming.

Contrary to popular belief, an Ice Age is not a continual deep freeze, but a period of constantly fluctuating climate conditions punctuated by periods of intense cold. The earliest millennia of the last ice age—the critical period when our remote ancestors first colonized Africa—are little known. The information gleaned from deep-sea cores and ice borings gives us a much clearer picture of ice-age climate after Earth's magnetic field, generated deep inside our planet, abruptly reversed some 780,000 years ago. (It has not reversed since.) Deep-sea cores from the Pacific Ocean reveal at least nine major glacial (icy) periods that have come and gone over the past 780,000 years, the most recent of them ending in abrupt and irregular global warming between

HOW WE KNOW

DEEP SEA AND ICE CORES

Layers of sediment build up over time on ocean beds, and annual layers of ice are added to polar caps. By extracting cores of ice or deep sea sediment and looking at the composition, scientists can build a picture of climate change. Increases in atmospheric CO₂ (carbon dioxide) and CH₄ (methane)—both greenhouse gases—can be detected in the ice and indicate warming.

Similarly, the ratios of oxygen isotopes in the shells of microscopic marine animals reflect changes in sea temperature. The Vostok ice core from Antarctica provides evidence for the last 420,000 years, and shows that major shifts in temperature occur about every 100,000 years.

